

Assignment 4: Literature Review & Analytical Questions (Zeng et al., 2019)

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Task: Literature survey, questions, and discussion on "Growth model interpretation of planet size distribution" (Li Zeng et al., 2019)

Discord Submission Status Note:

"Please note: Due to platform restrictions and access limitations in my region, I am unable to access the course Discord server. As discussed and approved, I am submitting my formulated research questions and analytical notes regarding the weekly paper directly through this document in our shared Google Drive directory and my GitHub repository."

Analytical Questions Derived from Zeng et al. (2019)

Question 1: Distinguishing Water Worlds from Gas Dwarfs via Interior Structure Models and Volatile Fraction Degeneracy

Question:

In the introduction and discussion sections, Zeng et al. (2019) address the long-standing compositional ambiguity of sub-Neptunes (2 to 4 Earth radii), noting that planets in this radius range can either be gas dwarfs (rocky cores with a thin hydrogen-helium envelope) or water worlds (significant amounts of water-dominated ices and fluids). The paper demonstrates that a planet with a mass of 10 to 15 Earth masses containing roughly equal parts ice and rock matches the second peak of the radius distribution.

My question is: Given that both gas dwarfs and water worlds require only a few mass percentage points of gas to match observed radii, how can future high-precision atmospheric and interior modeling constrain the boundary between superionic water phases and primordial hydrogen-helium envelopes when analyzing transit timing variation masses versus radial-velocity masses?

Analytical Answer & Discussion:

Resolving the composition degeneracy between a volatile-rich water world and a gas dwarf relies heavily on moving beyond simple mass-radius measurements and incorporating multi-wavelength atmospheric characterization and interior state equations of state.

- **Mean Molecular Weight and Atmospheric Signatures:**

- A primordial hydrogen-helium envelope has a very low mean molecular weight, which results in a large atmospheric scale height, making transmission spectroscopy features quite prominent. Conversely, an envelope or secondary atmosphere originating from a volatile-rich water interior will be heavily enriched in heavier molecules like water vapor, carbon dioxide, and methane, significantly lowering the scale height and dampening spectral amplitudes. Future instruments targeting transmission features look for high-altitude clouds and precise water-to-hydrogen abundance ratios to differentiate these profiles.

- **High-Pressure Equation of State Constraints:**

Deep in the interior of planets ranging from 2 to 4 Earth radii, pressures exceed tens of gigapascals, where water transitions into exotic phases such as superionic ice. These high-pressure phases alter the density profile significantly compared to a standard silicate core enveloped by a compressible ideal-gas hydrogen-helium layer.

- **Transit Timing Variation vs. Radial-Velocity Discrepancies:**

Transit timing variation-derived masses tend to systematically skew lower than radial-velocity masses for planets in similar radius bins, often correlating with stellar metallicity differences. Combining homogenous radial-velocity datasets with transit timing variations allows modelers to constrain tidal dissipation and density stratification, providing a clearer window into whether the outer envelope is a loosely bound volatile gas layer or a high-density fluid and superionic mantle.

Question 2: Evaluating the Photoevaporation Scenario versus Intrinsic Protoplanetary Disk Condensation Fronts

Question:

In analyzing the bimodal radius distribution and the local minimum gap at 1.8 to 2.0 Earth radii, the authors challenge the dominant photoevaporation and core-powered mass-loss mechanism. They point out that the log-uniform orbital distance distribution of both subpopulations beyond approximately 10 days conflicts with photoevaporation

models, which strongly depend on stellar insolation and orbital separation. Instead, they argue the gap stems from intrinsic properties and condensation sequences in the protoplanetary disk, such as the location of the water snowline.

My question is: If the radius gap is primarily an imprint of primordial disk composition and ice condensation fronts rather than secondary photoevaporation, how do we account for the observed dependence of the radius valley's location on stellar mass and host star metallicity in subsequent empirical surveys like the California-Kepler Survey?

Analytical Answer & Discussion:

The debate between primordial disk-assembly mechanisms and post-formation atmospheric stripping represents one of the most vibrant discussions in modern exoplanetology.

- **Stellar Mass and Insolation Scaling:**

Empirical studies from the California-Kepler Survey clearly show that the location of the radius valley slopes downward to smaller radii for lower-mass M-dwarfs compared to G-type stars. Purely primordial snowline models must reconcile how the stellar host mass dynamically shifts the thermal structure and condensation fronts of the protoplanetary disk in a way that matches this precise scaling.

- **The Hybrid Perspective:**

While Zeng et al. argue that the flatness of orbital distance distributions challenges pure photoevaporation models, modern consensus leans toward a nuanced interplay. Protoplanetary disks naturally establish distinct compositional groups based on condensation fronts, such as rock-dominated inner zones versus ice-rich outer zones via pebble accretion. When these embryos migrate inward, they establish the bimodal baseline. Subsequent secondary processes can further refine or carve out the valley, meaning the final architecture is likely a combined footprint of both formation location and subsequent atmospheric evolution.

- **Metallicity Dependence:**

Host star metallicity dictates the total solid-to-gas ratio in the disk. Metal-rich stars foster larger, more massive cores that can accrete and retain more substantial volatile or gaseous envelopes. Accounting for this requires integrating dynamic disk evolution models with stellar population synthesis frameworks.

Question 3: The Role of Primordial Core Collisions in Shaping the Mass Distribution of Ice Giants

Question:

When explaining the higher-mass peak of the radial-velocity sub-Neptune population around 16 Earth masses, Zeng et al. (2019) propose that these cores are formed through the collisional merging of multiple smaller, primordial icy cores rather than runaway gas accretion. The authors suggest that collisions between icy cores can partially strip their primordial envelopes while retaining core mass, unlike rocky cores above 10 Earth masses which tend to undergo catastrophic disruption.

My question is: What are the specific hydrodynamic and thermodynamic thresholds during giant impacts of icy embryos that dictate whether an accretionary collision results in successful core merger versus complete envelope vaporization, and how do these Monte Carlo collision models scale across different initial protoplanetary disk metallicities?

Analytical Answer & Discussion:

The proposition that Uranus, Neptune, and many super-Neptune exoplanets are products of multiple primordial icy core mergers rather than single isolated growth tracks provides a compelling solution to why they avoided runaway gas giant status despite exceeding critical core masses.

- **Impact Mechanics of Ice versus Rock:**

Collisional physics simulations demonstrate that high-energy impacts between massive rocky bodies often lead to catastrophic fragmentation due to the brittle, high-density nature of silicate-metal matrices. In contrast, icy cores containing significant volatile and fluid fractions exhibit different shock-absorption and plastic deformation properties. When two icy embryos collide, the low-density volatile envelopes can absorb and dissipate a fraction of the kinetic shock, allowing the dense ice-rock cores to coalesce while stripping away or shocking the surrounding primordial gas envelope.

- **Bypassing the Runaway Gas Threshold:**

Standard core-accretion theory dictates that once a core reaches roughly 10 Earth masses, it should undergo unchecked gas accretion into a gas giant. However, if the final mass is achieved late in the disk's lifetime via an energetic merger of two sub-critical cores, the local nebular gas may already be depleted or the impact itself may have dynamically blasted away the remaining envelope, leaving behind a massive water-rich ice giant.

- **Scaling with Disk Metallicity:**

In protoplanetary disks with higher solid surface densities and higher metallicity, the frequency of planetary embryo formation and subsequent scattering events increases dramatically. Consequently, Monte Carlo simulations that incorporate these collision-merger algorithms predict an elevated population of massive icy sub-Neptunes and water worlds in metal-rich environments, aligning well with observed exoplanet occurrence rates.